

Performance Evaluation of Passive Herringbone Micromixer

ME 224 - Fundamentals of Microscale Flows

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Abstract:

Rapid and efficient mixing of fluids at the microscale is a fundamental challenge in the development of Lab-on-a-Chip (LOC) technologies. As microfluidic flow operates at extremely low Reynolds numbers, the flow remains strictly laminar. Consequently, natural mixing relies entirely on molecular diffusion, which is a highly inefficient and slow process. To address this, passive micromixers utilize complex channel geometries to manipulate the flow without external energy sources. This project evaluates the performance of a Passive Herringbone Micromixer using Computational Fluid Dynamics (CFD). The herringbone design consists of staggered, asymmetric grooves to induce transverse pressure gradients, converting axial flow into helical vortices that stretch and fold the fluid interface through "chaotic advection".

In this study, we investigated the mixing performance of three geometric configurations: a 3-inlet flow-focusing geometry, a 2-inlet Y-shaped geometry, and a plane microchannel. Simulations were conducted using ANSYS to solve the Navier-Stokes and advection-diffusion equations for a water-ethanol mixture across five different inlet velocities. Mixing efficiency was evaluated using a variance-based Mixing Index. The results demonstrate that herringbone geometries drastically reduce the required mixing length compared to plain channels. The 3-inlet geometry outperformed the 2-inlet design due to its increased interfacial area. Importantly, we confirmed an inverse relationship between inlet velocity and mixing quality: lower velocities increase the fluid's residence time, allowing molecular diffusion to successfully homogenize the chaotically advected layers.

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1. Introduction

In Microchannels, fluid flow is strictly laminar (very low Reynolds Number), as there is a dominance of viscous forces over inertial forces due to the length scale and other factors. Under these conditions, the fluid flows in parallel layers without turbulence to blend them together. As there is no turbulence, mixing in microchannels happens through molecular diffusion, which is a process that is often too slow for practical, continuous-flow applications.

In passive mixers, no external energy sources are necessary. Passive micromixers rely on the channel's internal geometry to alter the flow. The Staggered Herringbone Micromixer (SHM) is a highly effective passive design. It features a sequence of slanted, asymmetric V-shaped grooves on the channel floor. As fluid flows over these grooves, it experiences transverse pressure gradients that induce a secondary helical motion. This "chaotic advection" physically stretches and folds the fluid layers over one another, exponentially increasing the interfacial area and drastically reducing the time and distance required for diffusion to complete the mixing process.

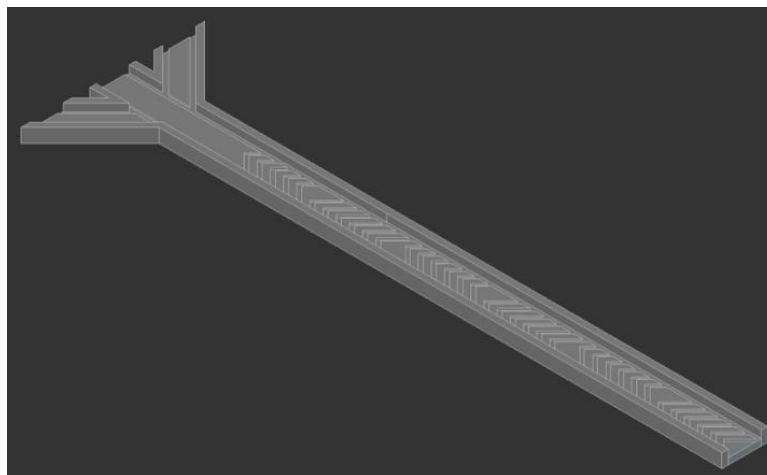


Fig 1: CAD Model of Herringbone Micromixer with 3 inlets

2. Literature Review

Reference Paper: A Numerical Research of Herringbone Passive Mixer at Low Reynold Number Regime - Wang et al., 2017

This paper used 3D Computational Fluid Dynamics (CFD) to analyze chaotic mixing mechanisms at extremely low Reynolds numbers.

Key Findings:

- The study compares mixing performances under two different driving forces (Pressure driven flow vs EOF).
- The study proved that in passive micromixers, lower flow rates actually yield higher mixing quality.
- A lower velocity increases the fluid's "residence time" inside the channel, allowing molecular diffusion more time to act across the fluid layers stretched by the herringbone grooves.
- Governing equations for laminar flow, species transport, and the variance-based statistical method used to calculate the dimensionless Mixing Index.

3. Motivation and Objective

Efficient micro-mixing is crucial for the functionality of Lab-on-a-Chip devices, biochemical assays, and drug delivery systems. However, introducing complex grooves into a microchannel increases hydrodynamic resistance, leading to a higher pressure drop. Therefore, designing an optimal mixer requires balancing high mixing quality against the pressure drop caused by the grooves.

The primary objectives of this project are:

- To simulate and evaluate a Passive Herringbone Micromixer using CFD.
- To compare mixing performance across three distinct geometries: a 3-Inlet (Flow Focusing) design, a 2-Inlet (Y-Shaped) design, and a Plane Microchannel with no grooves.
- To quantify the mixing index in each geometry for various flow rates and Reynolds Numbers.

4. Methodology

4.1. Geometric Modelling

The 3D CAD models for our microfluidic domains were generated using Fusion 360. Three geometries were modelled: the 3-Inlet flow-focusing geometry, the 2-Inlet Y-junction, and the control Plane geometry. The herringbone models featured repeating cycles of asymmetric, 45° slanted grooves patterned on the channel floor.

4.2. Simulation Setup

The geometries were imported into ANSYS. Because our flow operates strictly in the laminar regime, we solved the steady-state Navier-Stokes equations coupled with a Species Transport model. We assigned water as the bulk fluid and tracked ethanol as the diluted species to evaluate mixing.

4.3. Meshing and Boundary Conditions

To accurately capture the micro-vortices near the grooves, we applied a physics-controlled fine mesh. At the inlets, we assigned a mass fraction of 1 for the ethanol stream and 0 for the water stream. The simulations were run across five different inlet velocities to observe the impact of flow rate on performance.

4.4. Quantitative Evaluation

Instead of relying purely on visual contours, we extracted the standard deviation of the concentration at various cut-planes along the channel. We then calculated the dimensionless Mixing Index (MI), where an MI of 0 indicates completely unmixed streams and an MI of 1 indicates perfect homogeneity.

5. Results and Discussion

5.1. Performance of the 3-Inlet Geometry

We tested the 3-Inlet geometry across five varying inlet velocities.

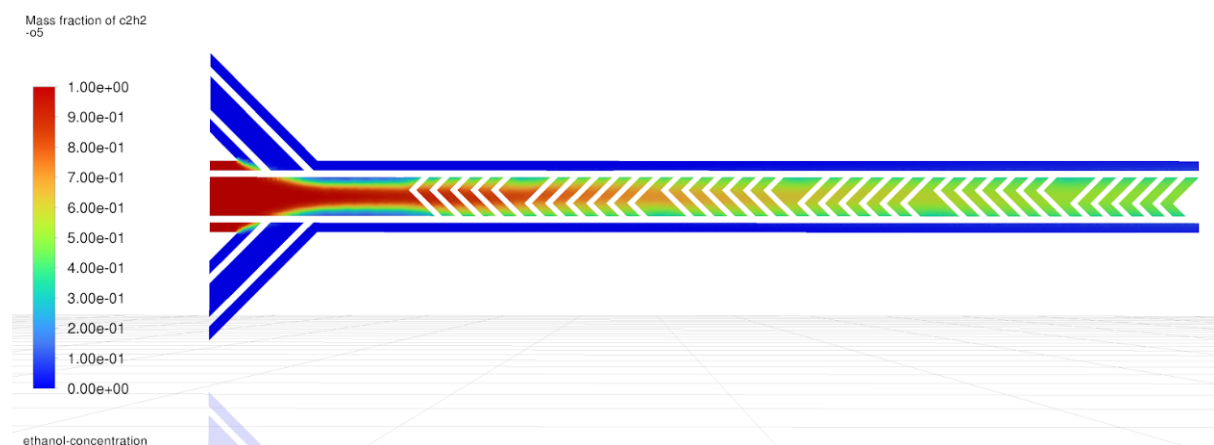


Fig 2: Mass fraction contour plot for 3-Inlet Geometry

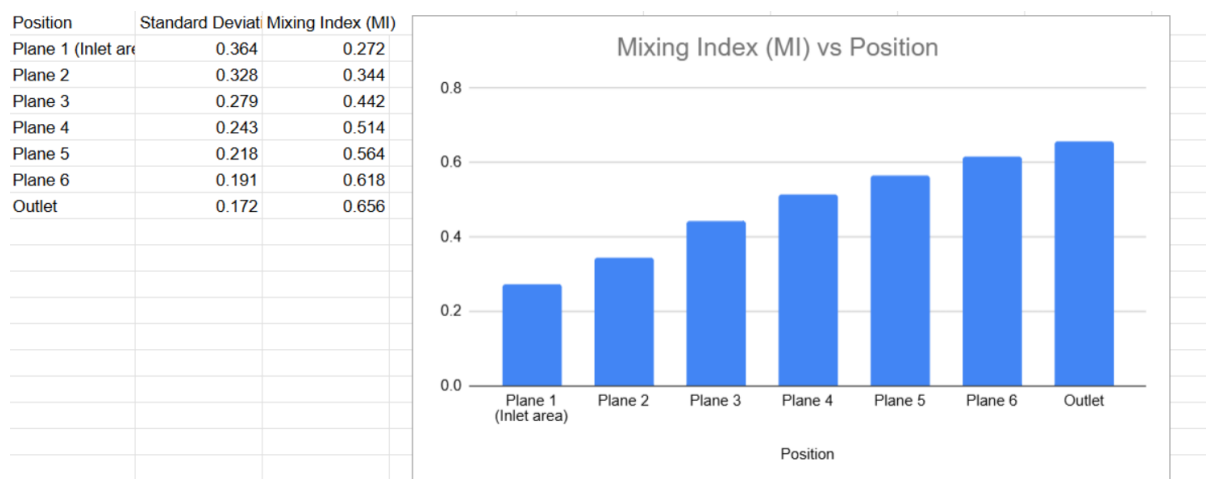


Fig 3: Bar Graph of Mixing Index vs Position for 3-Inlet

Case	Velocity (v)	Flow Rate (Q)	Area (A)	Reynold's Number (Re)	Mixing Index (Outlet)
Case 1	0.00085	0.6732	1.32×10^{-8}	0.0618	0.656
Case 2	0.0012	0.9504	1.32×10^{-8}	0.0872	0.6147
Case 3	0.00185	1.4652	1.32×10^{-8}	0.1345	0.583
Case 4	0.0001	0.0792	1.32×10^{-8}	0.0073	0.9836
Case 5	0.000085	0.06732	1.32×10^{-8}	0.0062	0.9947

Fig 4: Table containing Case 1 to 5 data for 3-Inlet Geometry

The contour plots clearly visualize chaotic advection breaking down the fluid stratification. When analyzing the data, a strong inverse relationship between velocity and mixing performance emerged. In Case 5, operating at a very low velocity (0.000085 m/s), the mixer achieved a near-perfect Mixing Index of 0.9947 at the outlet. However, at higher speeds like Case 3 (0.00185 m/s), the mixing index dropped to 0.583. This confirms that at higher velocities, the fluid does not have enough residence time for molecular diffusion to act across the folded fluid layers.

5.2. Performance of the 2-Inlet Geometry

A similar parametric sweep was conducted for the 2-Inlet Y-shaped geometry.

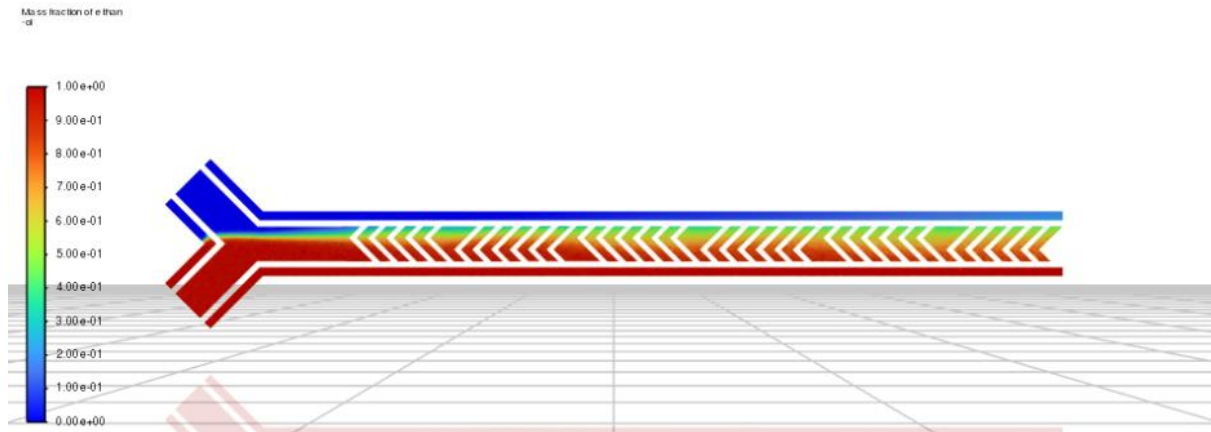


Fig 5: Mass fraction contour plot for 2-Inlet Geometry

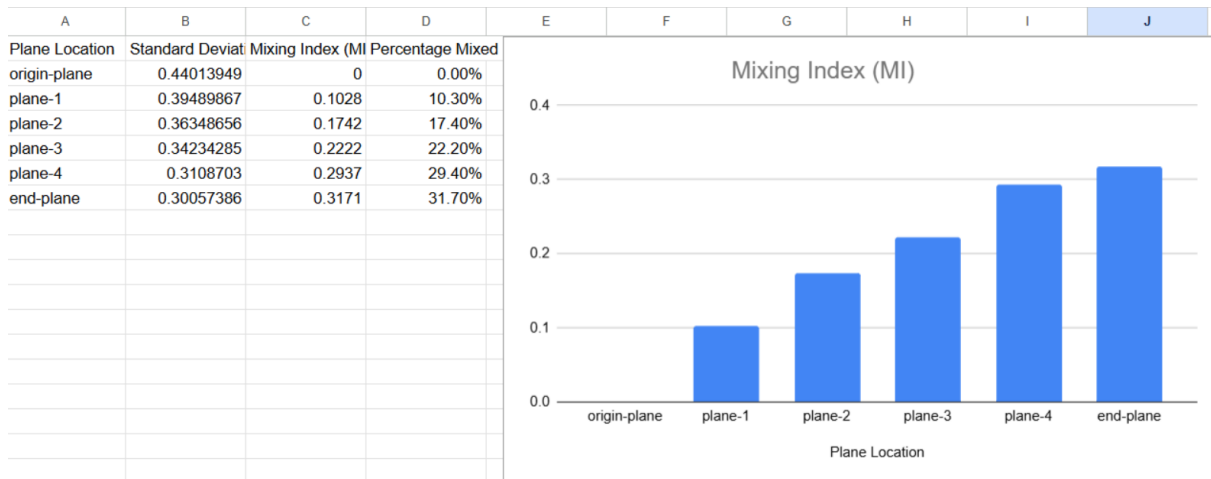


Fig 6: Bar Graph of Mixing Index vs Position for 2-Inlet Geometry

Case	Velocity (v)	Flow Rate (Q)	Area (A)	Reynold's Number (Re)	Mixing Index (Outlet)
Case 1	0.00085	0.6732	1.32 x 10 ⁻⁸	0.0618	0.656
Case 2	0.0012	0.9504	1.32 x 10 ⁻⁸	0.0872	0.6147
Case 3	0.00185	1.4652	1.32 x 10 ⁻⁸	0.1345	0.583
Case 4	0.0001	0.0792	1.32 x 10 ⁻⁸	0.0073	0.9836
Case 5	0.000085	0.06732	1.32 x 10 ⁻⁸	0.0062	0.9947

Fig 7: Table containing Case 1 to 5 data for 2-Inlet Geometry

The 2-Inlet geometry exhibited the same trend: decreasing the velocity increased the mixing index. Case 5 yielded an MI of 0.998, while Case 1 dropped to 0.345. However, when comparing the two designs, the 3-Inlet geometry consistently outperformed the 2-Inlet setup. This is because the 3-Inlet design sandwiches the solvent between two solute streams,

effectively doubling the initial interfacial contact area and giving the advection process a significant head-start.

5.3. Performance of Plane Geometry

To prove the necessity of the grooves, we ran a simulation on a plane microchannel at an inlet velocity of 0.0001 m/s.

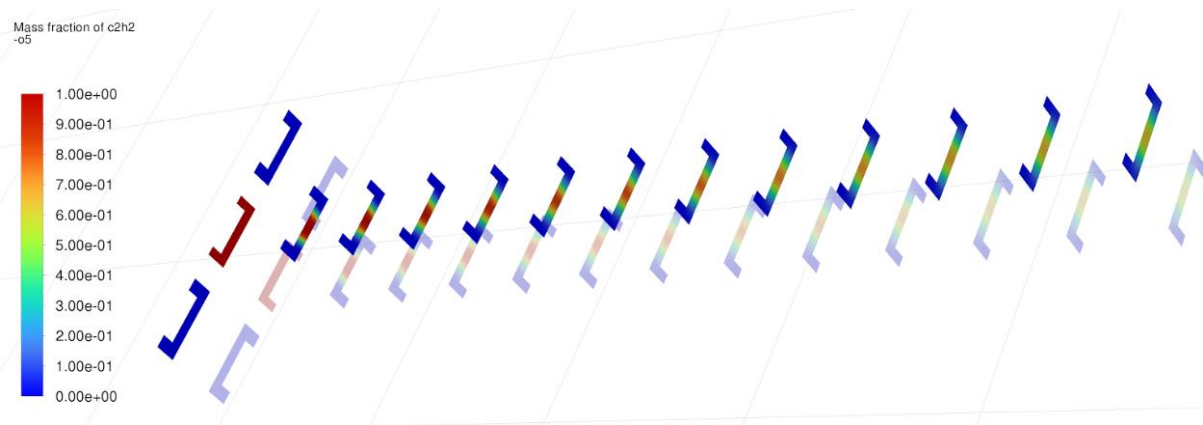


Fig 8: Mass fraction contour plot for Plane Channel

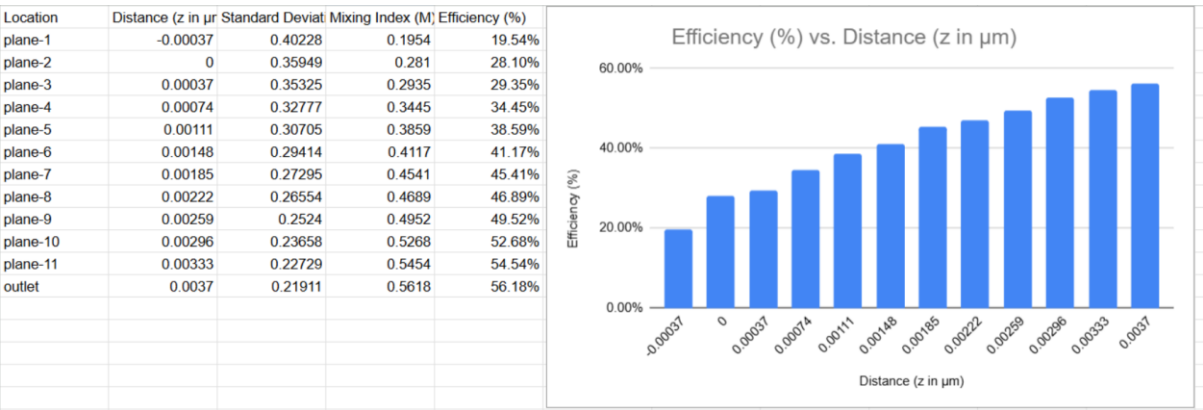


Fig 9: Bar Graph of Mixing Index vs Position for Plane Channel

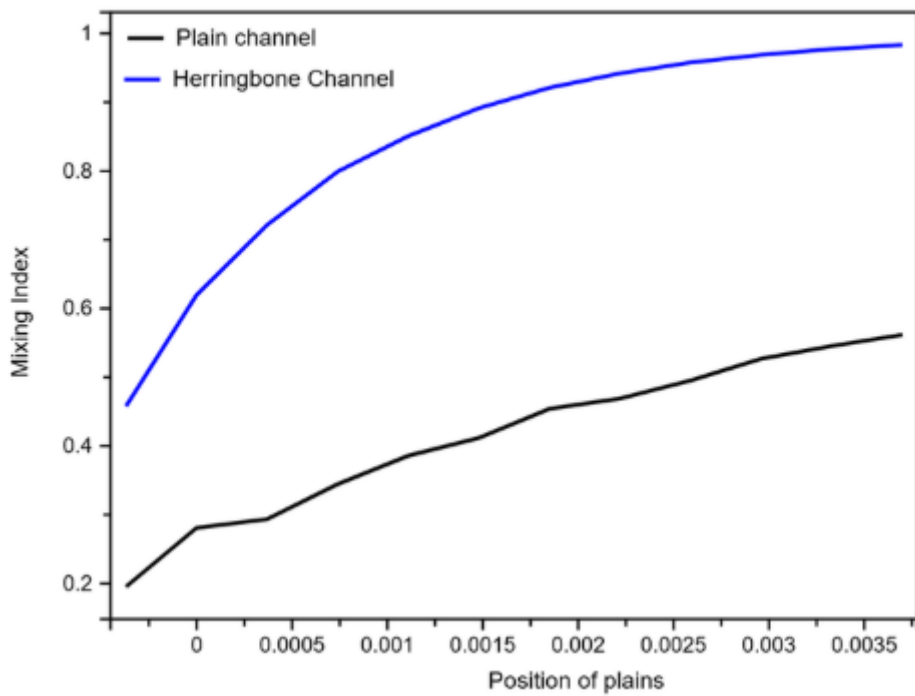


Fig 10: Line Graph comparing Mixing Index of Plain Channel vs Herringbone Channel

The parameters obtained are:

Velocity (v): 0.0001 m/s

Area (A): $1.32 \times 10^{-8} \text{ m}^2/\text{s}$

Flow Rate (Q): 0.0792 $\mu\text{L}/\text{min}$

Reynold's Number (Re): 0.0073

Mixing Index (At Outlet): 0.5618

Without the herringbone structures, the plane channel only reached a mixing index of 0.5618 at the outlet. Operating at the same velocity, the herringbone geometries achieved an index of roughly 0.98. The comparative line graph highlights this stark contrast: the plane channel's mixing progresses linearly and slowly via pure diffusion, while the herringbone channel rapidly homogenizes the fluid through chaotic advection.

6. Conclusions

Based on the results, we conclude that:

1. The herringbone geometry decreases the required mixing length compared to a plane microchannel. The mixing index at the outlet of the herringbone geometry is much higher than that of the plane microchannel.
2. Performance of the 3-inlet geometry was better than that of the 2-inlet geometry, as the interfacial area at the inlet is doubled in the case of the 3-inlet geometry.
3. As the velocity at the inlet increases, the mixing index at the outlet decreases, and as we decrease the velocity at the inlet, the mixing index at the outlet increases. Lowering the velocity increases the residence time, allowing molecular diffusion to happen for a longer period of time.

7. Future Scope

This baseline CFD study opens several avenues for future research:

Varying the Dimensions: Creating multiple geometries with different dimensions to identify and verify the optimal groove depth-to-height ratio and asymmetry index using CFD.

Non-Newtonian Fluids: Extending the simulation to include non-Newtonian fluid models (e.g., blood, polymer melts) for specific biomedical or industrial applications.

Experimental Validation: Fabricating the 3-Inlet design using PDMS soft-lithography and conducting fluorescent dye microscopy to validate the CFD data against a physical prototype.

References

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